

Slice sampling

The sampling identity, the algorithm, and why it works

Econometrics 32706

The problem

We want draws from a density known only up to a normalizing constant:

$$\pi(x) = \frac{q(x)}{Z}, \quad Z = \int_{\mathbb{R}} q(t) dt.$$

We can evaluate $q(x)$, but calculating Z or sampling directly from π may be difficult.

Key idea: represent π as the marginal distribution of a point chosen uniformly from a region under the graph of q . This representation leads to slice sampling.

Proposition: uniform sampling under a graph

Let $q : \mathbb{R} \rightarrow [0, \infty)$ be measurable and $0 < Z = \int_{\mathbb{R}} q(x) dx < \infty$. Define

$$A = \{(x, h) \in \mathbb{R} \times (0, \infty) : 0 < h < q(x)\}.$$

Sampling proposition

If (X, H) is uniform on A with respect to area, then X has density

$$\pi(x) = \frac{q(x)}{Z}.$$

The height H is auxiliary: discard it after sampling (X, H) , and the remaining X follows the target distribution.

Source: Neal (2003), *Slice sampling*, *Annals of Statistics* 31(3), 705–767, doi:10.1214/aos/1056562461.

Proof of the sampling proposition

Adding vertical strips of height $q(x)$ (Tonelli's theorem) gives

$$|A| = \int_{\mathbb{R}} \int_0^{q(x)} dh \, dx = \int_{\mathbb{R}} q(x) \, dx = Z.$$

Thus, the area of A is Z .

Define a uniform density on A :

$$p(x, h) = \frac{1}{|A|} \mathbf{1}\{(x, h) \in A\} = \frac{1}{Z} \mathbf{1}\{0 < h < q(x)\}.$$

Its marginal density for X is π :

$$p_X(x) = \int_0^{q(x)} \frac{1}{Z} \, dh = \frac{q(x)}{Z} = \pi(x).$$

Conclusion: sample (X, H) from this joint density and keep only X ; then $X \sim \pi$.

Conditional 1: the height given the position

From slide 4,

$$p(x, h) = \frac{1}{Z} \mathbf{1}\{0 < h < q(x)\}, \quad p_X(x) = \frac{q(x)}{Z}.$$

For a fixed x with $q(x) > 0$, divide the joint density by the marginal density of X :

$$\begin{aligned} p(h \mid x) &= \frac{p(x, h)}{p_X(x)} \\ &= \frac{1}{q(x)} \mathbf{1}\{0 < h < q(x)\}. \end{aligned}$$

This is constant on the vertical interval $(0, q(x))$, so

$$H \mid X = x \sim U(0, q(x)).$$

Why the height marginal contains $|S_h|$

Fix a height $h > 0$. The horizontal slice is $S_h = \{x : q(x) > h\}$; let $|S_h|$ be its total length.

At this height, the joint density is $1/Z$ for every $x \in S_h$, and zero for every $x \notin S_h$:

$$p(x, h) = \frac{1}{Z} \mathbf{1}\{x \in S_h\}.$$

To find the marginal density of H , add the joint density over all possible positions:

$$\begin{aligned} p_H(h) &= \int_{\mathbb{R}} p(x, h) dx = \int_{S_h} \frac{1}{Z} dx \\ &= \frac{1}{Z} |S_h|. \end{aligned}$$

Intuition: the longer the horizontal slice at h , the more area of A lies near that height.

Conditional 2: the position given the height

By definition $S_h = \{z : q(z) > h\}$. For a fixed $h > 0$,

$$x \in S_h \iff q(x) > h \iff 0 < h < q(x).$$

The two indicators therefore agree. The joint density on slide 4 can be written, for $h > 0$, as

$$p(x, h) = \frac{1}{Z} \mathbf{1}\{0 < h < q(x)\} = \frac{1}{Z} \mathbf{1}\{x \in S_h\}.$$

We also showed $p_H(h) = |S_h|/Z$ when $|S_h| > 0$. Divide the joint density by this height marginal:

$$\begin{aligned} p(x | h) &= \frac{p(x, h)}{p_H(h)} \\ &= \frac{1}{|S_h|} \mathbf{1}\{x \in S_h\}. \end{aligned}$$

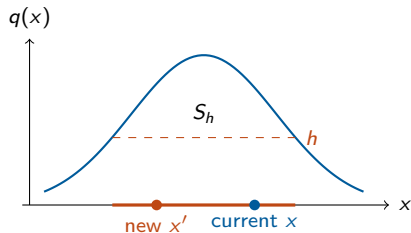
Thus every position in S_h has the same conditional density, and positions outside it have density zero:

$$X | H = h \sim U(S_h).$$

How to use it

The joint law has two conditional distributions:

$$h \mid x \sim U(0, q(x)), \quad x \mid h \sim U(S_h).$$



The slice S_h contains every position whose density exceeds the chosen height.

Alternate draws from $H \mid X$ and $X \mid H$. This constructs a Markov chain whose stationary joint law is uniform on A .

The ideal slice-sampling update

Starting from the current position x :

1. Draw $u \sim U(0, 1)$, then set $h = u q(x)$.
2. Find the horizontal slice $S_h = \{z : q(z) > h\}$.
3. Draw x' uniformly from S_h .
4. Use x' as the next state and repeat.

Since $h < q(x)$, the current position always lies in the slice. A new height is drawn on every update, so the set of possible moves changes with the current state.

A simple numerical example

Let $q(x) = e^{-x^2/2}$, the standard normal kernel, and suppose the current position is $x = 1$. If the random draw is $u = \frac{1}{2}$, then

$$h = u q(1) = \frac{1}{2} e^{-1/2}.$$

The slice condition is

$$e^{-z^2/2} > \frac{1}{2} e^{-1/2} \quad \Longleftrightarrow \quad |z| < \sqrt{1 + 2 \log 2} \approx 1.54.$$

Thus $S_h \approx (-1.54, 1.54)$, and the ideal update draws x' uniformly from this interval.

The example illustrates one possible height; the next update draws a new one.

Why the update targets π

Both steps sample from conditionals of the *same* joint law:

$$x \xrightarrow{h|x} (x, h) \xrightarrow{x'|h} (x', h).$$

If $x \sim \pi$, the first draw gives $(x, h) \sim p(x, h)$. The second conditional draw leaves that joint law unchanged. Discarding h therefore gives $x' \sim \pi$.

This is a Gibbs-sampling argument. It proves that π is stationary for the update; successive states need not be independent.

Another intuition: high-density points occur on many possible slices, so they receive more probability in the long run.

When the slice is hard to sample from

The ideal update assumes we can draw uniformly from S_h . For a complicated density, its endpoints may be unknown and S_h may have several pieces.

A practical one-dimensional slice sampler constructs a bracket around the current point, tests candidate positions, and shrinks the bracket after rejections. The bracket rules must preserve the correct conditional distribution on the slice.

The height step remains $h = u q(x)$. If we work with log densities, the equivalent threshold is

$$\log h = \log q(x) + \log u.$$

This form avoids evaluating a tiny density value directly.

What to remember

1. Uniform sampling under an unnormalized density curve gives the desired marginal distribution.
2. A random height selects a horizontal slice; the position update moves within that slice.
3. Alternating these conditional updates preserves the joint law and therefore the target marginal.

Reference: Radford M. Neal (2003), *Slice sampling*, *Annals of Statistics* 31(3), 705–767, doi:10.1214/aos/1056562461.